AI-Powered Banking Customer Complaint Management using BERT

A Mini Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINNERING

By

B. Kundana Mani – 2420030558

M. Sai Kridhay Kumar – 2420030211

Under the Esteemed Guidance of

Dr. K. Swanthana



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING



DECLARATION

The Mini Project Report entitled "AI-Powered Banking Customer Complaint Management using BERT" is a record of Bonafide work of B. Kundana Mani – 2420030558, M. Sai Kridhay Kumar – 2420030211 submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

B. Kundana Mani – 2420030558

M. Sai Kridhay Kumar – 2420030211

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING



CERTIFICATE

This is certify that the mini project-based report entitled "AI-Powered Banking Customer Complaint Management using BERT" is a Bonafide work done and submitted by B. Kundana Mani – 2420030558, M. Sai Kridhay Kumar – 2420030211 in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year 2026-2027.

Signature of the Course Co-ordinator **Signature of HOD**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, **Dr. K. Swanthana of Section 7**, Department of Computer Science which cannot be expressed in words for her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. P. Venkateshwar Sir, Head of the *Department for Computer Science Engineering*** for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

B. Kundana Mani – 2420030558

M. Sai Kridhay Kumar – 2420030211

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	
2.	Introduction	
3.	Literature survey	
4.	Problem Statement and Objectives	
5.	Hardware and Software requirements	
6.	Implementation	
7.	Results & Discussion	
8.	Conclusion & Future Scope	
9.	References	

1. ABSTRACT

Banking institutions receive large volumes of customer complaints related to accounts, credit cards, loans, transactions, and other financial services. Manually categorizing these complaints can be time-consuming and may lead to inconsistent classification. This project presents CORA (Complaint Orchestration and Routing Architecture for Automated Banking Complaint Management), a system designed to classify banking complaints and support structured complaint processing.

The project uses consumer complaint narratives derived from the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database. The proposed approach uses a pretrained Bidirectional Encoder Representations from Transformers (BERT) model, fine-tuned to classify complaints into Product, Sub-product, Issue, and Sub-issue. Preprocessing includes dataset inspection, handling missing and duplicate records, preparing target labels, and tokenizing complaint narratives. BERT provides contextual text representations, and the classifier is evaluated using Accuracy, Precision, Recall, and F1-Score.CORA proposes connecting classification outputs to predefined business rules for ticket generation, department mapping, priority assignment, and complaint tracking. This approach aims to support more structured banking complaint management.

Team Name: CORA

Team Members:

B. Kundana Mani, 2420030558 — Project concept, problem identification, dataset research, literature review, and methodology planning.

M. Sai Kridhay Kumar, 2420030211 — Project development and technical collaboration.

Specify his individual tasks according to the work he actually completed.

Dataset/source and project repository:

Dataset: CFPB Consumer Complaint Database

Dataset URL: https://www.consumerfinance.gov/data-research/consumer-complaints/

Project repository: https://www.github.com.KLH-CSE-2026-27-2420030558-ComplaintManagement

2. INTRODUCTION

2.1 Application Area

CORA (Complaint Orchestration and Routing Architecture for Automated Banking Complaint Management) focuses on automated banking customer complaint classification using Natural Language Processing (NLP) and Machine Learning. Banks and financial institutions receive complaints related to accounts, credit cards, loans, payments, transactions, and other financial services. These complaints are usually written in natural language and may differ in length, vocabulary, writing style, and level of detail. CORA aims to classify complaint narratives into structured categories to support organized complaint handling.

2.2 Need for Automated Text Analysis

Manually analysing a large volume of customer complaints is time-consuming and difficult to manage consistently. Different customers may describe similar problems using different words, while a single complaint may contain multiple details about a transaction or financial service. Manual classification can therefore require considerable effort and may result in inconsistent categorization or delays in processing complaints.

Machine Learning and NLP techniques can identify useful patterns in unstructured text and use those patterns to predict relevant categories. Contextual language models such as BERT can represent words based on their surrounding context, helping the system interpret complaint narratives beyond simple keyword matching. Automating the classification process can reduce repetitive manual work and provide structured information for further complaint processing.

2.3 Target Users and Intended Use

The intended users of CORA include bank complaint-handling teams, customer-support representatives, complaint management departments, and financial service organizations. The system is intended to assist these users by categorizing incoming complaints and providing structured outputs that can support subsequent routing and tracking. The predicted categories are intended to assist complaint processing rather than replace human review where it is required.

2.4 Input and Predicted Output

The input to CORA is a customer's written complaint narrative obtained from the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database. The system processes the text and predicts four category labels:

Product: The main financial product associated with the complaint.

Sub-product: A more specific category within the product.

Issue: The primary problem described by the customer.

Sub-issue: A more detailed classification of the problem.

These predictions provide a structured representation of the original complaint and can serve as inputs to a business-rule-based complaint management workflow.

2.5 Proposed Project Pipeline

The proposed CORA pipeline consists of the following stages:

- **Data collection:** Obtain consumer complaint narratives from the CFPB Consumer Complaint Database.
- **Data cleaning and preprocessing:** Inspect the dataset, identify missing and duplicate records, prepare the target labels, and organize the text and corresponding categories.
- **Feature representation:** Use the BERT tokenizer to convert complaint narratives into token IDs and attention masks suitable for model input.
- **Model training:** Fine-tune a pretrained BERT model with classification heads for Product, Sub-product, Issue, and Sub-issue prediction.
- **Model evaluation:** Evaluate the classification outputs using Accuracy, Precision, Recall, and F1-Score.
- **Complaint processing:** Use the predicted categories as inputs for the proposed business-rule stage, which may support ticket generation, department mapping, priority assignment, and tracking.

- **User interface:** A user interface may be used to submit complaints and display classification results and processing status, depending on the implemented application features.

2.6 Organization of the Report

The remaining chapters are organized as follows. Chapter 3 presents the literature survey and compares existing approaches to text classification and complaint analysis. Chapter 4 defines the problem statement and project objectives. Chapter 5 describes the hardware and software requirements. Chapter 6 explains the implementation methodology, dataset preparation, model architecture, training, and evaluation process. Chapter 7 presents the experimental results and discusses the model's performance and limitations. Finally, Chapter 8 summarizes the conclusions and future enhancements, while Chapter 9 lists the references used in the project

3. LITERATURE SURVEY

Research on customer complaint analysis has explored topic modelling, intent classification, contextual language models, and automated complaint categorization. These approaches provide a foundation for CORA, which focuses on classifying banking complaint narratives into structured categories and connecting the predictions to a complaint-management workflow.

3.1 Review of Existing Research

3.1.1 Bastani et al. (2019) — LDA for CFPB Consumer Complaints

Bastani et al. (2019) investigated how topic modelling could reveal recurring themes in consumer financial complaints. The study used complaint narratives from the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database. The authors applied Latent Dirichlet Allocation (LDA) to discover topics and support complaint analysis. The resulting topics provided insights into common financial issues and consumer concerns. However, topic discovery does not directly predict predefined complaint labels or implement an end-to-end complaint-routing process [1].

3.1.2 Devlin et al. (2019) — BERT for Contextual Text Representation

Devlin et al. (2019) introduced Bidirectional Encoder Representations from Transformers (BERT) to improve contextual language understanding. The work evaluated BERT on established natural-language understanding benchmarks rather than a dedicated banking complaint-management dataset. BERT learns representations using both left and right textual context and can be fine-tuned for downstream classification tasks. The authors reported strong results across multiple language-understanding benchmarks, demonstrating the usefulness of contextual pretraining. Nevertheless, the paper does not address banking-specific complaint categories, business-rule routing, or complaint lifecycle management [2].

3.1.3 Casanueva et al. (2020) — Banking Intent Classification

Casanueva et al. (2020) studied efficient intent detection for conversational systems operating in domains with limited labelled examples. Their work introduced BANKING77, a dataset containing 13,083 annotated banking queries distributed across 77 intent categories. The authors evaluated pretrained dual sentence encoders, including Universal Sentence Encoder and ConveRT,

against alternative intent-detection approaches. The encoders performed competitively, particularly in few-shot settings, and could be trained efficiently with limited computational resources. However, banking-query intent detection differs from detailed financial complaint classification and does not itself manage complaint tickets or routing [3].

3.1.4 Grootendorst (2022) — BERTopic for Topic Discovery

Grootendorst (2022) proposed BERTopic, a neural topic-modelling approach designed to discover meaningful themes in collections of documents. The method uses pretrained transformer-based document embeddings, clustering, and a class-based TF-IDF procedure to represent the discovered topics. The study presented and evaluated BERTopic as an alternative to traditional topic-modelling approaches. Contextual embeddings and clustering can help produce meaningful topic representations. However, BERTopic is primarily an unsupervised topic-discovery method; it does not directly provide CORA's four predefined complaint labels or an integrated complaint-management workflow [4].

3.1.5 Roumeliotis et al. (2025) — Reasoning LLMs for Complaint Classification

Roumeliotis et al. (2025) examined whether large language models and reasoning models could classify financial complaints without task-specific fine-tuning. The study used CFPB consumer complaints and consolidated the original product categories into five broader financial classes. Fourteen language and reasoning models were evaluated using complaint text, with Accuracy, Precision, Recall, F1-score, and confusion-matrix analysis. The study investigated the effectiveness of these models in assigning broad financial complaint categories, although performance varied across models and overlapping classes remained challenging. The approach focuses on zero-shot classification rather than fine-tuning a dedicated model to predict four complaint attributes and integrating those predictions into a complaint-routing workflow [5].

3.1.6 Shin et al. (2026) — LLM Use in Consumer Financial Complaints

Shin et al. (2026) investigated the adoption and effectiveness of large language models in the context of US consumer financial complaints. The study analyzed CFPB complaint narratives to examine the relationship between LLM-assisted complaint writing and consumer outcomes. The researchers used empirical analyses to investigate patterns in LLM adoption and the relationship

between assisted complaints and consumer relief. Their findings suggest that LLM-assisted complaint writing may help consumers present their concerns and may be associated with differences in complaint outcomes. However, the study examines complaint writing and consumer outcomes rather than implementing CORA's supervised four-category classification and operational ticket-management pipeline [6].

3.2 Comparison of Existing Approaches

The reviewed studies address different aspects of text analytics. LDA and BERTopic discover recurring complaint themes, while BERT provides contextual representations that can be adapted to supervised classification. BANKING77 research demonstrates how banking-related text can be classified into specific intents, and recent LLM studies explore classification or the broader use of language models in financial complaints.

These approaches differ from CORA in their primary objectives. Topic modelling identifies themes without necessarily assigning predefined labels, whereas intent classification focuses on the purpose of a customer query. Contextual language models support detailed text classification, but classification alone does not automatically create tickets, assign departments, determine priorities, or support complaint tracking.

3.3 Research Gap

The reviewed literature establishes the value of topic modelling, contextual text representations, and automated classification for analyzing customer-written text. However, these studies do not, individually, demonstrate the complete CORA workflow of predicting Product, Sub-product, Issue, and Sub-issue from banking complaint narratives and connecting the predictions to a structured complaint-management process.

CORA addresses this project-level gap by combining a fine-tuned BERT classifier with a proposed business-rule stage for ticket generation, department mapping, priority assignment, employee allocation, and complaint tracking. The classifier is evaluated separately using Accuracy, Precision, Recall, and F1-score. These components represent the project's intended integration; improvements in routing efficiency or complaint-resolution time must be established through separate system-level evaluation rather than assumed from text-classification results alone.

4. PROBLEM STATEMENT AND OBJECTIVES

4.1 Problem Statement

To develop a BERT-based banking complaint classification system that analyses customer complaint narratives and predicts the corresponding Product, Sub-product, Issue, and Sub-issue categories to support structured complaint management and routing.

4.2 Problem Description

Banks receive large volumes of customer complaints related to financial products and services, making manual classification time-consuming and prone to inconsistent categorization. Traditional keyword-based methods may fail to capture the contextual meaning of complaint narratives, potentially leading to incorrect classification and routing. CORA uses a fine-tuned BERT model to classify complaint text into four categories, with the intended users being banking complaint-handling teams and customer-support personnel. The predicted categories can support subsequent business-rule-based routing, prioritization, ticket generation, and complaint tracking.

4.3 Objectives

The objectives of CORA are to:

- Obtain and inspect the CFPB Consumer Complaint Database, documenting its source, relevant fields, dataset size, and distribution of target categories.
- Clean and preprocess complaint narratives by inspecting missing and duplicate records and preparing the target labels for model training.
- Convert complaint text into numerical representations using the pretrained BERT tokenizer, generating token IDs and attention masks.
- Fine-tune a pretrained BERT model with classification heads to predict Product, Sub-product, Issue, and Sub-issue from complaint narratives.
- Evaluate the classification model using Accuracy, Precision, Recall, and F1-score for each prediction category.
- Connect the predicted complaint categories to the proposed business-rule workflow for department mapping, priority assignment, ticket generation, employee allocation, and complaint tracking, where these application components are implemented.

5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 Hardware Requirements

- **Processor:** Computer or laptop with a modern multi-core CPU.
- **RAM:** Minimum 8 GB RAM; 16 GB recommended for BERT model training.
- **Internet:** Required to download datasets, pretrained models, and software packages.
- **GPU:** Optional; recommended for faster BERT model training.
- **Storage:** Sufficient disk space for datasets, libraries, saved models, source code, and result figures.

5.2 Software Requirements

- **Programming Language:** Python
- **Development Environment:** Visual Studio Code
- **Data Processing Libraries:** Pandas and NumPy.
- **Machine Learning and Evaluation:** Scikit-learn.
- **Deep Learning Framework:** PyTorch.
- **NLP Library:** Hugging Face Transformers for BERT and tokenization.
- **Data Visualization:** Matplotlib and Seaborn
- **Dataset:** CFPB Consumer Complaint Database in CSV format, available at https://www.consumerfinance.gov/data-research/consumer-complaints/.
- **Version Control:** Git for tracking source-code changes.

6. IMPLEMENTATION

6.1 Data and Task

CORA (Complaint Orchestration and Routing Architecture) is an AI-powered banking complaint classification system designed to categorize customer complaints based on their textual narratives. The system uses the complaint narrative as input and predicts four target labels: Product, Sub-product, Issue, and Sub-issue. The dataset was obtained from the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database, which contains consumer complaints about financial products and services.

The project uses two CSV files: cora_train.csv, containing 10,157 records, and cora_validation.csv, containing 2,540 records. The dataset includes six Product categories, 31 Sub-product categories, 46 Issue categories, and 174 Sub-issue categories. The training dataset is used to train the model, while the validation dataset is used to evaluate its performance during development. A separate test dataset has not been documented. The dataset source is available at https://www.consumerfinance.gov/data-research/consumer-complaints/. The applicable CFPB data-use conditions should be reviewed before redistributing the dataset or derived data.

6.2 NLP Preprocessing and Feature Representation

The dataset is inspected to identify missing values, duplicate records, and inconsistencies in the complaint narratives and target labels. The text and corresponding category labels are prepared before model training. Since the selected model is BERT, traditional feature extraction methods such as Bag of Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF) are not used as the primary text representation.

The project uses the pretrained bert-base-uncased tokenizer to convert complaint narratives into token IDs and attention masks. The maximum sequence length is set to 128 tokens. The uncased tokenizer handles text casing as part of tokenization. Unlike conventional NLP pipelines, the BERT preprocessing approach does not require a separate stemming or lemmatization step, and stop-word removal is not applied as a separate preprocessing operation. This preserves the natural context of the complaint narrative for the transformer model.

6.3 Model Training and Evaluation

CORA uses a pretrained BERT model fine-tuned for multi-output complaint classification. A shared BERT encoder generates contextual representations of the complaint narrative, and four classification heads predict the Product, Sub-product, Issue, and Sub-issue labels independently. This design allows the model to learn contextual information from the complaint text while supporting four related classification tasks.

The model is implemented using PyTorch and the Hugging Face Transformers library. The AdamW optimizer is used with a learning rate of 2×10^{-5}. Training is performed with a batch size of 4 for one epoch using class-weighted cross-entropy loss. The class weights help address differences in the frequency of target categories. Training is performed using the CPU, without requiring a dedicated GPU.

Model performance is assessed using accuracy, precision, recall, and F1-score for each target label. These metrics provide complementary information about classification performance, particularly because the dataset contains different numbers of categories and may have an uneven distribution of examples across classes. Validation results are used to examine the model's performance across the four classification tasks. A separate random seed and a formal best-model selection procedure have not been documented and should not be reported as completed unless they were used in the implementation.

6.4 Project Workflow and Modules

The CORA implementation follows a sequence of data preparation, text preprocessing, model training, and evaluation.

Data loading and exploratory analysis: Loads the training and validation CSV files, examines the dataset structure, and identifies missing values, duplicates, and target-label distributions.

Text cleaning and preprocessing: Prepares complaint narratives and category labels and converts the text into token IDs and attention masks using the BERT tokenizer.

Feature representation and model training: Loads the pretrained BERT model, configures the four classification heads, and fine-tunes the model using the training data.

Evaluation and error analysis: Calculates accuracy, precision, recall, and F1-score for the four target labels and examines differences in performance across classification levels.

The project code and dataset files are organized under the CORA project repository, KLH-CSE-2026-27-2420030558-ComplaintManagement. The exact source filenames for these modules should be added from the actual repository structure rather than assumed.

6.5 Machine-Learning Implementation

The primary model selected for CORA is BERT, a transformer-based language model that uses contextual representations to understand the meaning of words within complaint narratives. Unlike conventional models that rely on manually constructed text features, BERT generates contextual representations that can capture relationships between words and phrases. This is useful for banking complaints, where similar wording may refer to different products or issues depending on the surrounding context.

The model uses one shared encoder and four classification heads to predict the target labels. PyTorch provides the training framework, while Hugging Face Transformers supplies the pretrained model and tokenizer. The model is optimized using AdamW and class-weighted cross-entropy loss. The implemented model is evaluated on the validation dataset using standard classification metrics. Traditional algorithms such as Multinomial Naive Bayes, Logistic Regression, Linear SVM, Decision Tree, and Random Forest are not listed as implemented comparison models because their use has not been confirmed in the project.

6.6 Demonstration and Application Workflow

The intended CORA application workflow begins with a customer submitting a banking complaint narrative. The trained BERT model predicts the corresponding Product, Sub-product, Issue, and Sub-issue categories. These predictions can then support a business-rule workflow for organizing complaints, creating tickets, mapping complaints to relevant departments, assigning priorities, and tracking their status.

The model training and evaluation workflow is documented as the core implementation. A completed Streamlit, Flask, or FastAPI interface, database integration, input-validation mechanism, and live ticket-assignment system should be described as implemented components only if they are present in the final application. If a demonstration interface has not been completed, the model can be demonstrated through a notebook or command-line workflow that accepts a complaint narrative and displays the predicted categories.

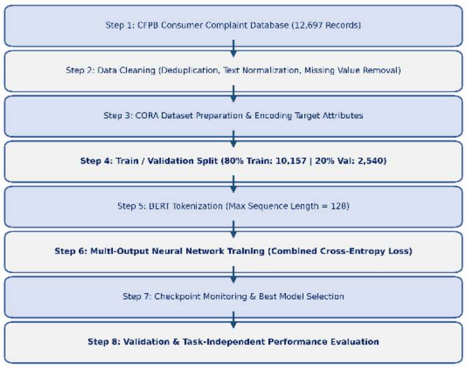
7. RESULTS & DISCUSSION

7.1 Dataset Summary

The CORA model was developed using the CFPB Consumer Complaint Database. The experimental dataset contains 12,697 records, divided into 10,157 training records (80%) and 2,540 validation records (20%). Each record contains a customer complaint narrative and its associated classification labels. The model predicts four attributes: Product, Sub-product, Issue, and Sub-issue, with 6, 31, 46, and 174 categories, respectively.

The increasing number of categories across the four classification tasks makes the prediction problem progressively more challenging. Product classification involves a relatively small number of broad categories, whereas Sub-issue classification requires the model to distinguish between a much larger number of specific complaint types. This difference is important when interpreting the evaluation results.

7.2 Experimental Workflow

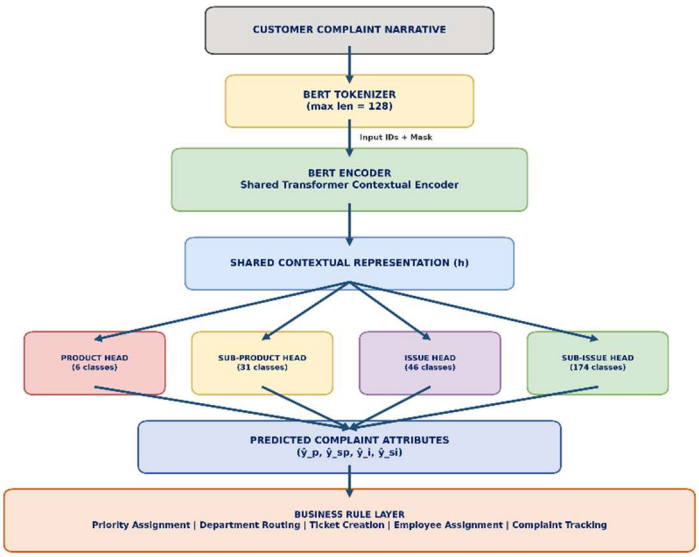


The experimental workflow begins with the CFPB complaint dataset and proceeds through data cleaning, target-attribute preparation, an 80:20 training-validation split, BERT tokenization, and multi-output model training. The maximum token sequence length is set to 128, and the trained model is evaluated separately for each classification task. The workflow also illustrates checkpoint monitoring and model selection; these steps should be described as completed only if they were performed in the actual training code.

7.3 CORA Model Architecture

The architecture uses a shared BERT encoder to generate contextual representations of each complaint narrative. These representations are passed to four classification heads, which

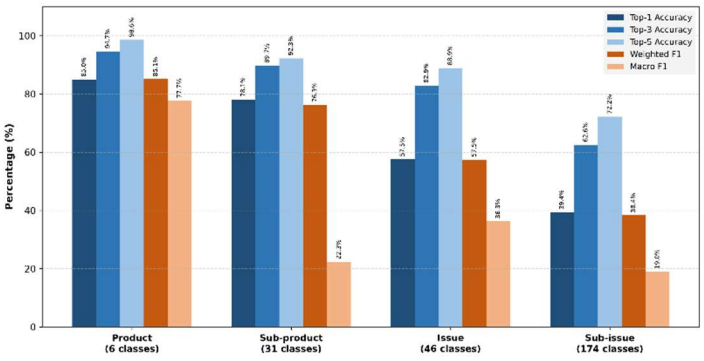
independently predict Product, Sub-product, Issue, and Sub-issue. The resulting predictions can support downstream business-rule operations such as priority assignment, department routing, ticket creation, employee assignment, and complaint tracking, provided that application components are implemented.



7.4 Classification Performance

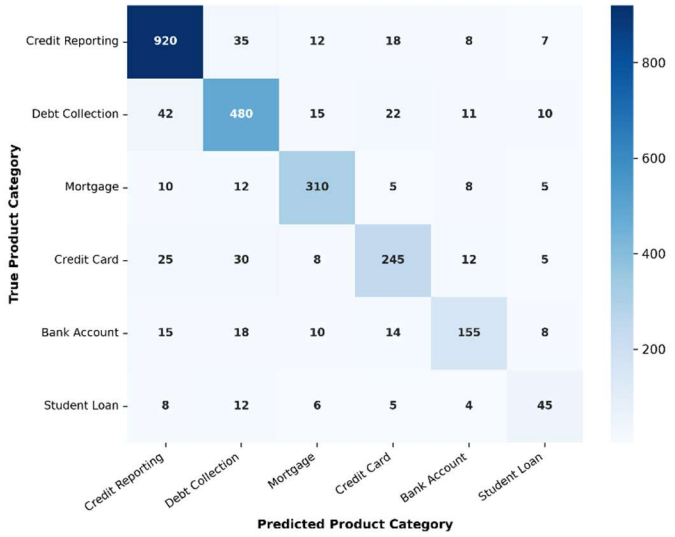
The performance chart compares Top-1, Top-3, and Top-5 accuracy, weighted F1-score, and macro F1-score across the four classification tasks. The reported Top-1 accuracy is 85.0% for Product, 78.1% for Sub-product, 57.5% for Issue, and 39.4% for Sub-issue. This trend indicates that the model performs better on broad complaint categories than on highly specific categories with many possible labels.

The chart also shows differences between weighted and macro F1-scores, particularly for Sub-product, Issue, and Sub-issue. Weighted F1 gives more influence to categories with more examples, whereas macro F1 gives equal importance to each category. The lower macro F1 values suggest that performance across individual classes may be uneven. This is consistent with the challenges of class imbalance and fine-grained classification, although the actual class distribution and per-class metrics should be examined to identify the specific causes.



7.5 Confusion Matrix Analysis

The confusion matrix compares the true Product categories with the categories predicted by CORA. The diagonal cells represent correct predictions, while the off-diagonal cells indicate misclassifications. The matrix shows strong diagonal counts for Credit Reporting and Debt Collection, indicating that many examples from these categories were correctly classified. Some errors occur between Credit Reporting and Debt Collection, as well as between Credit Card and Debt Collection, suggesting that certain complaint narratives may share similar language or characteristics.



The confusion matrix is useful for identifying categories that are frequently confused and for guiding further model improvements. However, this figure covers only the Product classification task. Separate confusion matrices or class-wise precision, recall, and F1-score results would be needed to analyze errors in Sub-product, Issue, and Sub-issue classification.

7.6 Discussion of Findings

The results indicate that the shared BERT encoder can classify complaint narratives across multiple related attributes, with stronger performance on the broader Product and Sub-product tasks than on the more detailed Issue and Sub-issue tasks. The Sub-issue task is particularly challenging because it contains 174 categories, making it harder to distinguish between closely related complaint types. The gap between weighted and macro F1 also highlights the importance of evaluating performance beyond overall accuracy.

Potential sources of error include overlapping complaint descriptions, limited training examples for some categories, and ambiguity in the language used by customers. Further investigation of class frequencies and misclassified examples would help determine the relative contribution of these factors. Possible improvements include collecting more examples for underrepresented

classes, tuning training parameters, evaluating alternative class-weighting strategies, and conducting systematic error analysis.

The model's predictions may support automated complaint organization and downstream routing. However, the quality of those operational decisions depends on the reliability of the predicted labels and the correctness of the business rules. Therefore, automated routing should be evaluated separately from the classification model before the complete system is considered ready for operational use.

7.7 Limitations

The reported results are based on a defined dataset split and should not automatically be interpreted as evidence of performance on every banking institution or complaint channel. The model is also limited by the available label quality, category imbalance, and the maximum input sequence length of 128 tokens. Longer complaint narratives may be truncated during tokenization, potentially removing relevant information.

The figures provided focus on classification metrics and the Product confusion matrix. Additional evidence, including the complete per-class evaluation results, the confusion matrices for the remaining tasks, and a comparison against baseline models, would provide a more comprehensive assessment. ROC curves should only be included if appropriate scores and a clearly defined one-vs-rest or other suitable evaluation method are available for the multiclass tasks.

7.8 Summary

The experimental results demonstrate that CORA can classify customer complaint narratives into four banking complaint attributes using a shared BERT encoder and separate classification heads. Performance is strongest for broad Product categories and declines for more granular labels, particularly Sub-issue. These findings support the potential of contextual language models for complaint classification while highlighting the need for further improvements in fine-grained prediction, class-wise performance, and downstream routing validation.

8. CONCLUSION & FUTURE SCOPE

8.1 Conclusion

CORA (Complaint Orchestration and Routing Architecture) was developed to address the challenges of manually categorizing banking customer complaints. The project uses the CFPB Consumer Complaint Database and a fine-tuned BERT model to classify complaint narratives into four attributes: Product, Sub-product, Issue, and Sub-issue. A shared BERT encoder with four classification heads was implemented using PyTorch and Hugging Face Transformers, with a maximum sequence length of 128 tokens.

The model was evaluated using accuracy, precision, recall, and F1-score. The experimental results show that classification performance is stronger for broader categories and decreases for more specific categories, particularly Sub-issue, which contains 174 classes. This highlights the difficulty of fine-grained complaint classification and the importance of considering class imbalance and differences in category complexity when evaluating the model.

The primary objective of developing a BERT-based complaint classification model was addressed through dataset preparation, tokenization, model training, and evaluation. The objective of examining classification performance was also addressed using the selected evaluation metrics. The proposed integration of predicted complaint attributes with business rules for priority assignment, department routing, ticket creation, employee assignment, and complaint tracking provides a direction for automating complaint management. However, the completion of these application components depends on the implementation of the corresponding interface and operational modules.

The most important limitation is the reduced performance on fine-grained complaint categories. The model's predictions are also influenced by the quality and distribution of the available training data. Overall, CORA demonstrates the potential of transformer-based NLP for structuring banking complaints, while further work is needed to improve detailed classification and validate the complete complaint-management workflow.

8.2 Future Scope

Improved class-imbalance handling: Explore alternative class-weighting strategies, resampling techniques, and additional training examples for underrepresented complaint categories.

Model optimization: Experiment with training parameters, sequence lengths, and suitable transformer variants to improve classification performance, particularly for Issue and Sub-issue.

Larger and more diverse datasets: Evaluate the model on additional complaint datasets and data from different financial institutions to assess its generalization ability.

Explainable predictions: Investigate methods for identifying the words or phrases that influence predicted complaint categories, helping users understand and review model outputs.

Application deployment: Integrate the trained classifier with a complaint-management interface and implement the proposed business rules for ticket creation, priority assignment, department routing, and complaint tracking.

Performance and latency testing: Measure prediction time, memory usage, and system responsiveness to determine whether the model is suitable for practical complaint-processing workloads.

Multilingual complaint classification: Extend the system to support complaints written in additional languages, subject to the availability of suitable multilingual training data and models.

These extensions could improve CORA's classification reliability, interpretability, and practical usability in banking complaint-management systems.

9. REFERENCES

Research Papers

[1] Bastani, K., Namavari, H., and Shaffer, J. (2019). "Latent Dirichlet Allocation (LDA) for Topic Modeling of the CFPB Consumer Complaints." *Expert Systems with Applications*, 127, 256–271. https://doi.org/10.1016/j.eswa.2019.03.001

[2] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 4171–4186. https://doi.org/10.18653/v1/N19-1423

[3] Casanueva, I., Temčinas, T., Gerz, D., Henderson, M., and Vulić, I. (2020). "Efficient Intent Detection with Dual Sentence Encoders." *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, pp. 38–45. https://doi.org/10.18653/v1/2020.nlp4convai-1.5

[4] Wolf, T., Debut, L., Sanh, V., et al. (2020). "Transformers: State-of-the-Art Natural Language Processing." *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pp. 38–45. https://doi.org/10.18653/v1/2020.emnlp-demos.6

[5] Grootendorst, M. (2022). "BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure." *arXiv preprint arXiv:2203.05794*. https://doi.org/10.48550/arXiv.2203.05794

[6] "Effective Complaint Detection in Financial Services through Complaint, Severity, Emotion and Sentiment Analysis." (2025). *Procedia Computer Science*, 258, 2220–2231. https://doi.org/10.1016/j.procs.2025.04.472

Dataset and Software Documentation

[7] Consumer Financial Protection Bureau (CFPB). *Consumer Complaint Database*. Available: https://www.consumerfinance.gov/data-research/consumer-complaints/.

[8] PyTorch Contributors. *PyTorch Documentation*. Available: https://pytorch.org/docs/stable/.

[9] Hugging Face. *Transformers Documentation*. Available: https://huggingface.co/docs/transformers/.

[10] scikit-learn Developers. *scikit-learn Documentation*. Available: https://scikit-learn.org/stable/.

[11] The pandas Development Team. *pandas Documentation*. Available: https://pandas.pydata.org/docs/.

[12] NumPy Developers. *NumPy Documentation*. Available: https://numpy.org/doc/.

[13] Matplotlib Development Team. *Matplotlib Documentation*. Available: https://matplotlib.org/stable/.

[14] Seaborn Developers. *Seaborn Documentation*. Available: https://seaborn.pydata.org/.